



**SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY**

INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

**DRAFT - Addressing Volumetric Bias in
Multi-Instance Semantic Segmentation
Using Voronoi Tesselation**

Sebastian Oßner



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Adressierung volumetrischer Befangenheit in
Multi-Instanz-Semantischer Segmentierung
durch Voronoi-Tessellation

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Submission Date:	DRAFT - 30.04.2026

Disclaimer

I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, 30.04.2026

Sebastian Oßner

Abstract

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1 Introduction

 **Problem Statement**  **Motivation**  **Reserach Question**

2 Background

This section is dedicated to the foundational knowledge required for the understanding and evaluation of the concepts and themes explored in this thesis. It will aim to preserve the most common definitions in the field as they currently stand.

2.1 Addressing the Instance Imbalance Problem

Instance imbalance refers to the issue of certain objects of which there are multiple present in a scene being represented different from other objects of the same type.

These objects are commonly called instances and are of particular interest in certain problems in object detection and segmentation, since many types of objects manifest as spatially separate and morphologically varied. Spatial separation is of particular note, as the content of this work does not concern itself with and should not be confused with instance segmentation. Instance segmentation identifies fundamentally separate objects from spatially overlapping views.

In multi-instance semantic segmentation, no such overlap is considered. Spatial separation is the definition of object differentiation. The eponymous multiple instances stem from the specific problem domain wherein views can be said to generally contain many of these spatially separate instances. Biomedical imaging is a domain which contains many such problem cases. Where anatomies or pathologies manifest as many spatially separate instances. Examples of instance segmentation and multi-instance semantic segmentation can be seen in  **Add figure with cellular instance segmentation and mitochondria as multiple instance semantic segmentation**

Particular problems arise when considering such cases, where instance size disparities and other variations can influence computational algorithms in ways that are not always true to clinical reality. In the case of many cancers for example, lesion size does not always correspond to clinical relevance and smaller cancer lesions can be malignant while larger ones might be benign. In such cases, assigning a higher importance to larger instances, whether it be implicit or explicit, can lead to undesirable outcomes.

This thesis aims to address and explore this issue, evaluating different approaches of compensating for biases in the computed importance of different instances.

(M. F. Rachmadi et al. 2024; Florian Kofler, Shit, et al. 2023; Zhang et al. 2021; F. Rachmadi et al. 2024)

2.2 Connected Components

Connected components are a well-established notion in topology and computer vision and a way to differentiate between non-empty disjointed subsets of a space. These subsets in our case are instances determined by a binary label map and a connectivity parameter. In the application of pixelated images, interconnectedness of components is determined by the neighborhood of a pixel within the component.

Connectivity, formal definition

Different connectivity parameters need to be chosen based on the dimension of the space (2D vs. 3D). In this work, 2D components are connected using 8-connectivity, while 3D components are determined using 26-connectivity. This is aligned with related works.

2.3 Voronoi Tesselation

The Voronoi tessellation algorithm is a fundamental building block of the approach outlined in this thesis. It describes a method of partitioning a space into regions based on certain points within the space.

This algorithm is especially suited to multiple spatially separate instances which are used to define the regions and their borders.

In the simple case of the euclidean plane, a finite set of 2D points $P = \{p_1, p_2, \dots, p_n\}, \subseteq \mathbb{R}^2$ must be given to “seed” the tessellation. Each point p_i is assigned a separate region (also called cell) $R_i \subseteq \mathbb{R}^2$. All other points in the plane are assigned the region that minimizes some distance function d :

$$R_k = \{x \in \mathbb{R}^2 \mid d(x, p_k) \leq d(x, p_j), \forall j \neq k\} \quad (1)$$

This definition can be analogously expanded to higher dimensional metric spaces.

In our case of image segmentation, there are not single seed points, but rather seed instances, the boundaries of which are the points on which the distances are calculated. Any point within the instance trivially has a distance of 0 to the nearest instance. Points outside the instance are assigned based on their distance to the nearest instance boundary.

Figure: An example of Voronoi partitioning in 2D and 3D

Formalize this and describe the 3D case and instances instead of points

3 Related Work

3.1 A Taxonomy of Losses

(Ma et al. 2021)

3.2 Instance-wise Losses

(Zhang et al. 2021; F. Rachmadi et al. 2024; M. F. Rachmadi et al. 2024; Florian Kofler, Shit, et al. 2023; Bouteille et al. 2025)  **Extend loss o**

3.3 Weight Maps

(Shirokikh et al. 2020)

4 Methodology

4.1 Datasets

The practical implementation of this thesis was evaluated against multiple datasets that span dimensionality (2D as well as 3D), modality (MRIs, electron microscopy) and various inherent anatomical features and pathologies that result in varied segmentation instance properties.

Due to the diverse nature of the underlying data, it was imperative to gather dataset metrics that encapsulate these varied instance properties not only to properly evaluate our fundamental hypotheses, but also to deal with noise and errors during training. In order to investigate why certain approaches worked better than others, concrete information on the size, morphology, distribution, and number of instances must be reported and taken into consideration before conclusions can be drawn. This subsection contains a description of the datasets used, their properties and calculated metrics as well as a comprehensive overview on the estimation of their fidelity that could serve as a basis for future research in multi-instance segmentation.

4.1.1 Metrics and Fidelity of Multi-Instance Segmentation Datasets

Since this thesis concerns itself with binary semantic segmentation, all datasets can be abstracted into their constituent components as follows: An image of shape $h * w$ ($h * w * d$ in the case of 3D) and a binarized label file of the same shape as its corresponding image. The label file is therefore partitioned into background pixels/voxels and $n \geq 0$ foreground pixels/voxels. The n foreground pixels/voxels then undergo a process of labeling based on their local neighborhood, in which they are split into $m \geq 0$ foreground instances (typically $n \gg m$). In 2D, this is based on 8-connectivity while 3D instances are labeled using the higher-dimensional equivalent of 26-connectivity.

The resulting instances have inherent properties that are important to examine both when formulating hypotheses as well as investigating noise and segmentation errors. As these properties are of such importance, prior works in the fields of [Medical Imaging Quality Assessment \(MIQA\)](#) and [Topological Data Analysis \(TDA\)](#) have provided comprehensive frameworks for identifying the properties of segmentation data and how these impact performance reporting ([Florian Kofler, Möller, et al. 2023](#); [Maier-Hein et al. 2022](#)). While these works place substantial focus on the selection and calculation of quantitative segmentation metrics, this section provides a more holistic interpretation of multi-instance dataset attributes that shall aid in the interpretation of results.

 **Description of removal of small annotation errors and what theoretical effects these instances would have and a link to the evaluation of intact vs cleaned datasets**

4.1.2 Brain Metastases

 **Split description, instance distribution, sizes and variance, etc.**

4.1.3 White Matter Hyperintensities

 **Split description, instance distribution, sizes and variance, etc.**

4.1.4 Platelet Organelles

 **Split description, instance distribution, sizes and variance, etc.**

4.1.5 Mitochondria

 **Split description, instance distribution, sizes and variance, etc.**

4.2 Segmentation Evaluation Metrics

Many works have previously discussed the importance of the choice of metrics and the need to adapt to the specific task at hand, (Maier-Hein et al. 2022) have provided concrete guidance in the choice of instance-wise metrics in segmentation problems and (Florian Kofler, Möller, et al. 2023) provide a tool to calculate these metrics.

Additionally, (Jaus et al. 2025) proposed a family of metrics that are of particular interest to us since they also use voronoi tessellation and aggregate metrics on each region separately to address instance volume imbalance during evaluation.

This subsection will provide a comprehensive overview of the metrics of interest, the rationales behind this choice and supplementary information on how predicted segmentations were evaluated.

 **Relatively metrics-agnostic implementation: No checkpointing, training to completion, mirror sentiment from nnUNet**

4.2.1 Global Metrics

Standard Dice, F2

4.2.2 Instance-wise Metrics

Instance F1, Instance Dice, CCDice, Instance Recall by volume

4.3 Loss Formulation and Weighted Combination

 **Tested Losses, Global vs. Local splits and weight distribution and normalization as potential shortcoming of previous studies** Why does everyone combine local and global losses? Why didn't anyone account for potential absolute value inconsistency when doing so?

4.4 Weight Maps

 **A description and formulation of tested weight maps and their impact on segmentation results** A special note of instance cleaning and weight maps inverse weighting as previously implemented. Voronoi-based maps as novel and precomputed due to image patching

4.5 Model Architecture

4.5.1 Adaptive U-Net

 **A description of a somewhat adaptive UNet that can handle 2D and 3D data**

4.5.2 Precomputation and Image Patching

 **Instance information, voronoi maps, and weight maps can be precomputed and patched alongside images and labels.** This can lead to artifacts in train patches that are devoid of instances, but are still tessellated as a region and therefore punish FPs more harshly than in LTLC.

5 Results

 An instance-based analysis of segmentation, analyse instance metadata like morphology and voronoi regions in results

 Dataset noise and coeff. of covariance per dataset and how the random seed in deterministic training influences results

 Global vs instance vs mixed with a focus on global vs mixed due to earlier established literature

 Global vs. local weight distribution

 Analysis of runtimes and relative efficiency

 Weight maps

 The effect of false instance removal on partitioning and weight maps in 2D

 Adaptive Weighting

6 Discussion

7 Conclusion

Appendix A: Supplementary Material

– Supplementary Material –

 **Your appendix content goes here**

Glossary

MIQA – Medical Imaging Quality Assessment [5](#)

TDA – Topological Data Analysis [5](#)

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